

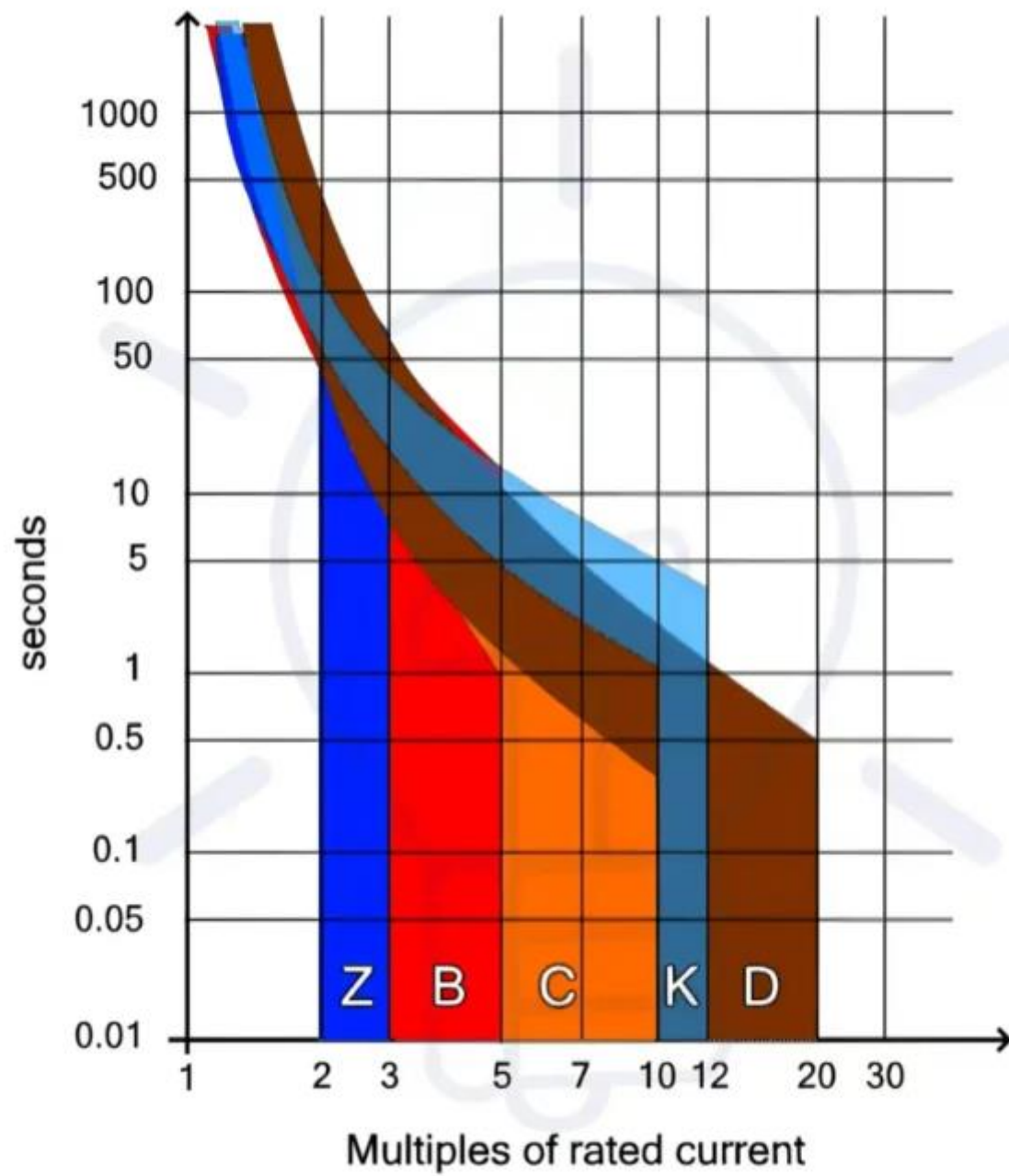
Tripping Curve

Lecture 10

Types of Circuit Breaker based on its Tripping Curve

- A tripping curve shows how quickly a miniature circuit breaker reacts to different levels of electrical current. The curve uses a graph with two axes. **The horizontal axis shows multiples of the breaker's rated current.** This means the graph does not use a unit like amps but instead shows how many times the normal current is flowing. **The vertical axis shows the time in seconds, using a logarithmic scale.** This helps people see how fast the breaker trips at different current levels.

The curve has two main parts. The first part is the thermal trip region. In this area, the breaker responds to small overloads that last longer. The curve here is smooth and shows that higher currents make the breaker trip faster. The second part is the magnetic trip region. This part reacts to very high currents, like a short circuit. The breaker trips almost instantly in this region.



Circuit breakers can be classified into different types based on their tripping characteristics and instantaneous tripping current.

Type A Miniature Circuit Breakers

Type A miniature circuit breakers are the most sensitive. They trip when the current is about double the rated value. Electricians use [Type A breakers](#) to protect sensitive electronics like semiconductors. These devices can break from even small extra currents. Type A breakers are not used much today. Most homes and businesses do not have Type A breakers.

***Note:** Type A breakers act fast but are not seen often in normal electrical systems. Fuses usually take their place for this job.*

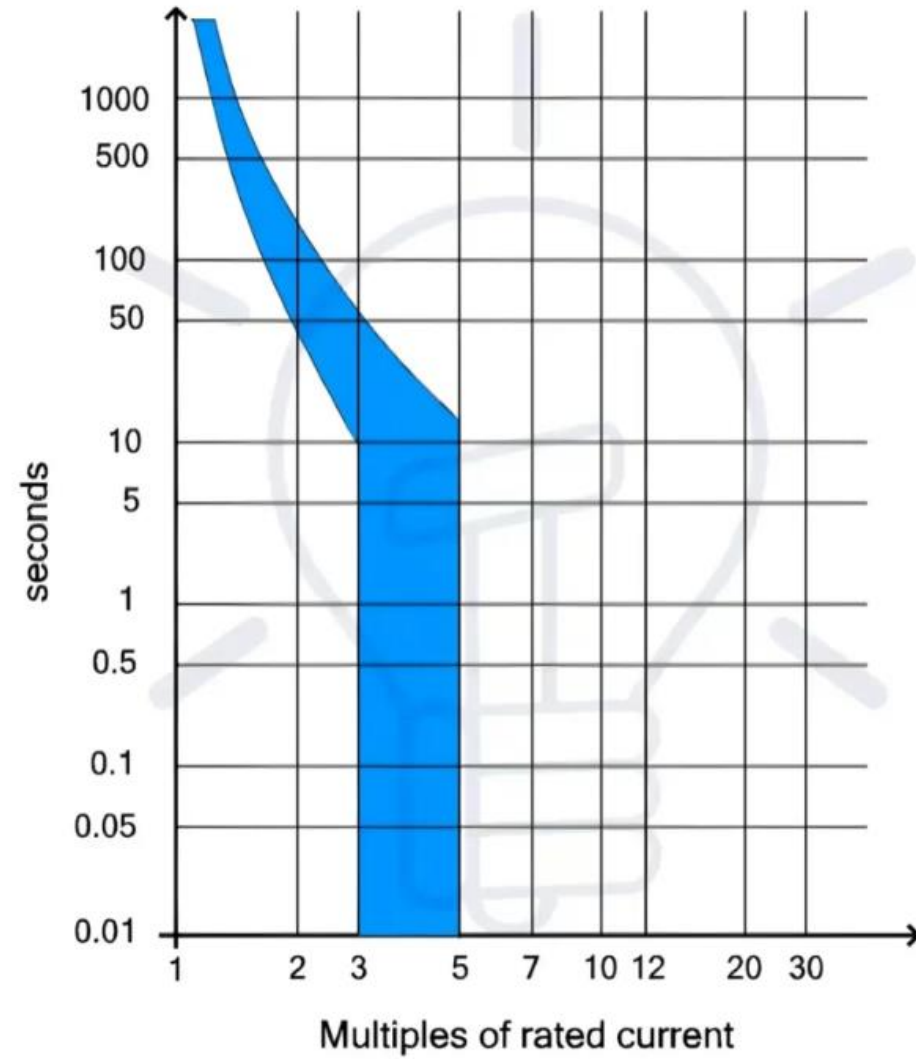
Type B Circuit Breaker:

This type of MCB instantly trips at 3 to 5 (2 to 3) times its rated current.

They are sensitive than other types and must not be used in places having higher surges than the mentioned limit. Otherwise, it will frequently trip circuit even in normal conditions. It can tolerate low surge current. They are suitable **for resistive loads in residential applications such as lighting and other resistive elements.**

Type B breakers trip quickly when there is high current. This helps keep sensitive equipment safe.

Type B Curve



Type C Miniature Circuit Breakers

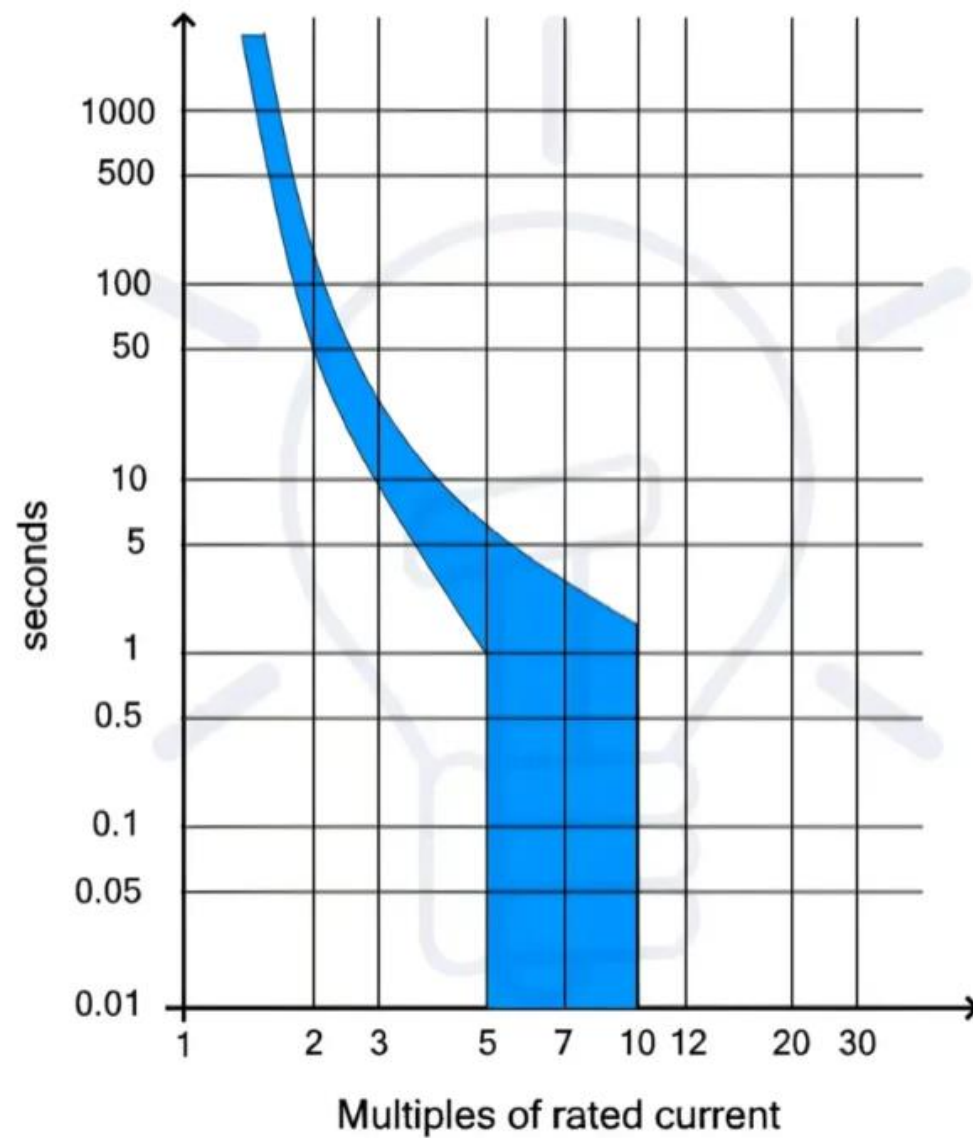
Such type of MCB instantly trips at 5 to 10 times its rated current. it is used for high inductive loads having high inrush current such as small **motors**. They are suitable for handling inductive loads in industries.

Type C breakers work with both resistive and inductive loads. This includes lights with bigger currents and small motors.

The thermal part bends a strip when there is a small overload. This takes 1 to 10 seconds, based on the current. The magnetic part acts much faster.

When a high current flows, the plunger trips the breaker in about 10 milliseconds. This lets Type C breakers stop both slow and sudden extra currents. They give a good mix of sensitivity and reliability.

Type C Curve

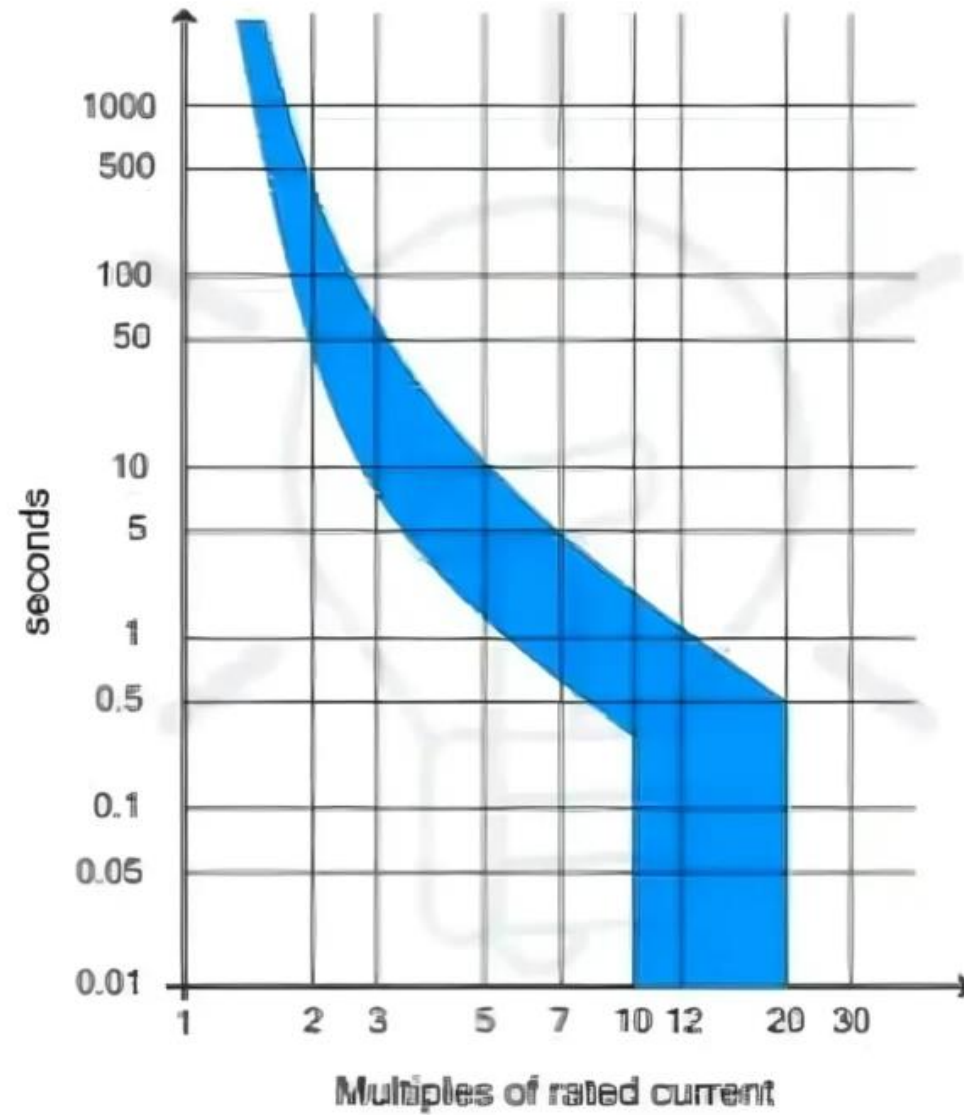


Type D Miniature Circuit Breakers

Such type of MCB trips instantly at 10 to 20 times its rated current. It can tolerate the max amount of surge a MCB can for a short duration. They are used for very high inductive loads such as heavy electrical motors in industrial and commercial applications.

These breakers are best for factories and big machines. Type D breakers are used for loads with high inrush currents. This includes large motors, transformers, and welding machines. Type D breakers let these machines start without tripping the circuit. But they still protect against dangerous faults. Factories often need Type D breakers because of heavy equipment.

Type D Curve



A table below shows the main features of each breaker type:

MCB Type	Trip Current Multiple	Typical Use Case	Notes
Type A	2 times rated current	Semiconductor protection (rarely used)	Usually replaced by fuses
Type B	2-3 times rated current	Purely resistive loads, low-voltage lighting, residential	Less commonly used currently
Type C	5-10 times rated current	Distribution lines, lighting circuits with larger currents	Most common for general use
Type D	10-20 times rated current	Systems with large inductive loads, high inrush currents	Industrial environments

Picking the right circuit breaker type is important. It depends on how much current the equipment uses. It also depends on how fast the breaker should trip. Each type has its own tripping curve. This means each one reacts to overloads and short circuits in a special way.

The following table lets you compare Type A, B, C, and D [miniature circuit breakers](#):

Breaker Type	Tripping Current (Times Rated)	Sensitivity	Typical Applications	Frequency of Use
Type A	2	Very High	Sensitive electronics, semiconductors	Rare
Type B	2-3	High	Lighting, small appliances, homes	Less common
Type C	5-10	Medium	Offices, schools, mixed loads	Most common
Type D	10-20	Low	Motors, transformers, factories	Industrial use

Type K Circuit Breaker

Such type of CB instantly trips at 8 to 12 times its rated current. they are used for heavy inductive loads in industries.

Type Z Circuit Breaker

Such type of circuit breaker instantly breaks the circuit at 2 to 3 times its rated current. they are the most sensitive CBs, and they are used for sensitive medical and semiconductor-based equipment that is exposed to small surges.

